Data preparation

- Pre-processing and data cleaning (Standard scaling, categorisation etc.) and eliminating redundancy.
- We prepare the data from baseline and extract trends from time-series values.
- A total of 24 features, 117 patients.
- 99 show no adverse reaction (IEC_HS=0) while 18 are impacted (IEC_HS=1).
- We find a total of 45 missing values over 30 patients.

Why XGBoost ?

- The key considerations are:
- 1) Sample efficiency.
- 2) No imputation/interpolation of the total of 45 missing values in 33 patients.
- Why no imputation/interpolation ?
- A) Clinically dubious. Cannot ‘assume’ or ‘extend’ patient data based on statistical analyses.
- B) Less number of data points- Even more unreliable after stratified train-test split.

Gradient Boosting

- So we need our supervised binary classifier to handle missing values natively as well as be reasonably accurate(overall and in terms of recall).
- Gradient boosting frameworks (such as XGBoost, Catboost,LightGBM) build an ensemble of decision trees sequentially minimizing the error of the previous.
- Hence XGBoost (out of 3 main contenders).

Best result (First Pass)

Overall accuracy: ~84.6% TP:6,TN:93,FP:6,FN:12

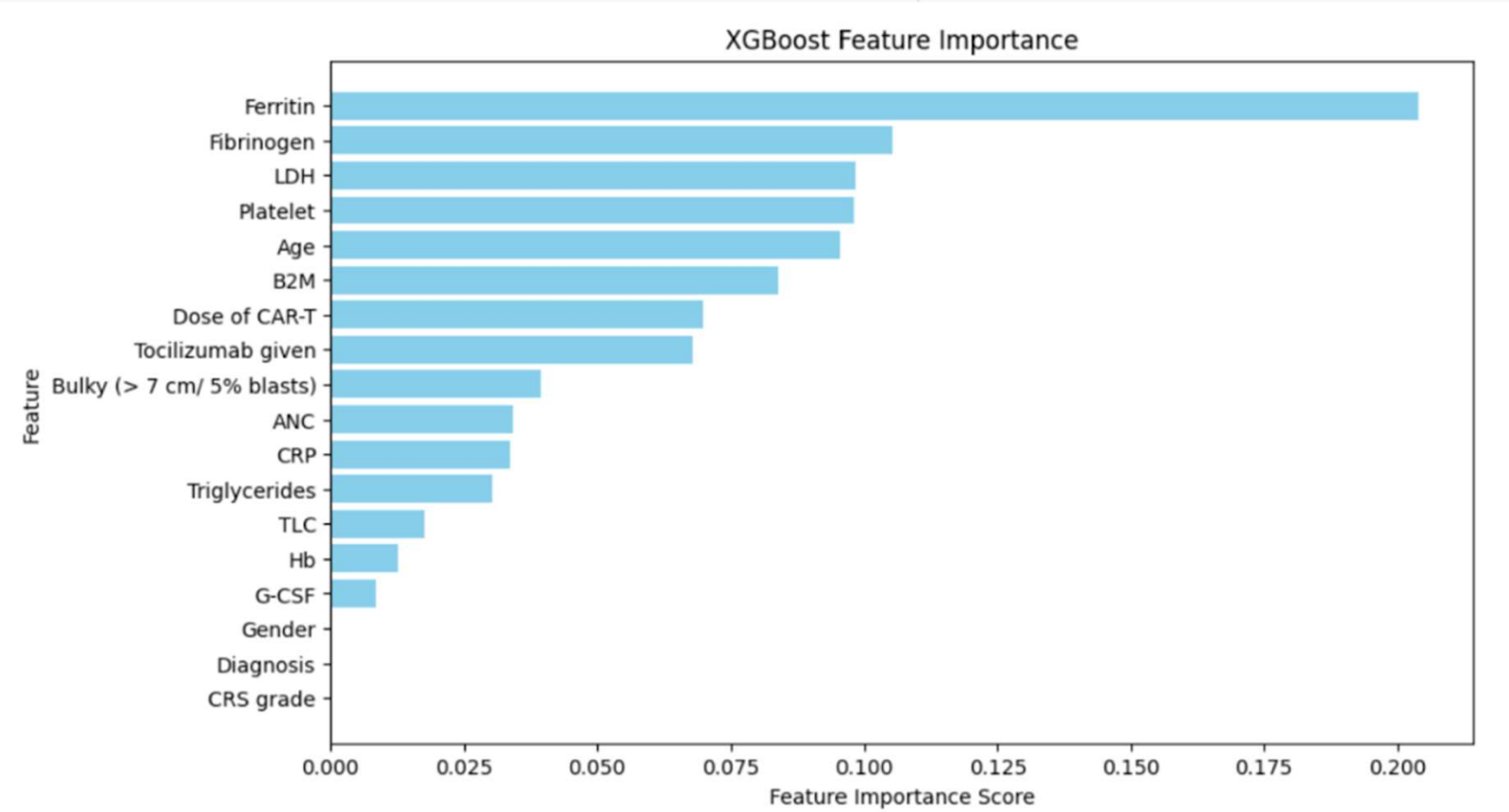
Precision: 50%, Recall: 33%(very poor), F1 Score: 40%

(XGBoost LOOCV with optimal hyperparameter set via optuna and manual tweaking)

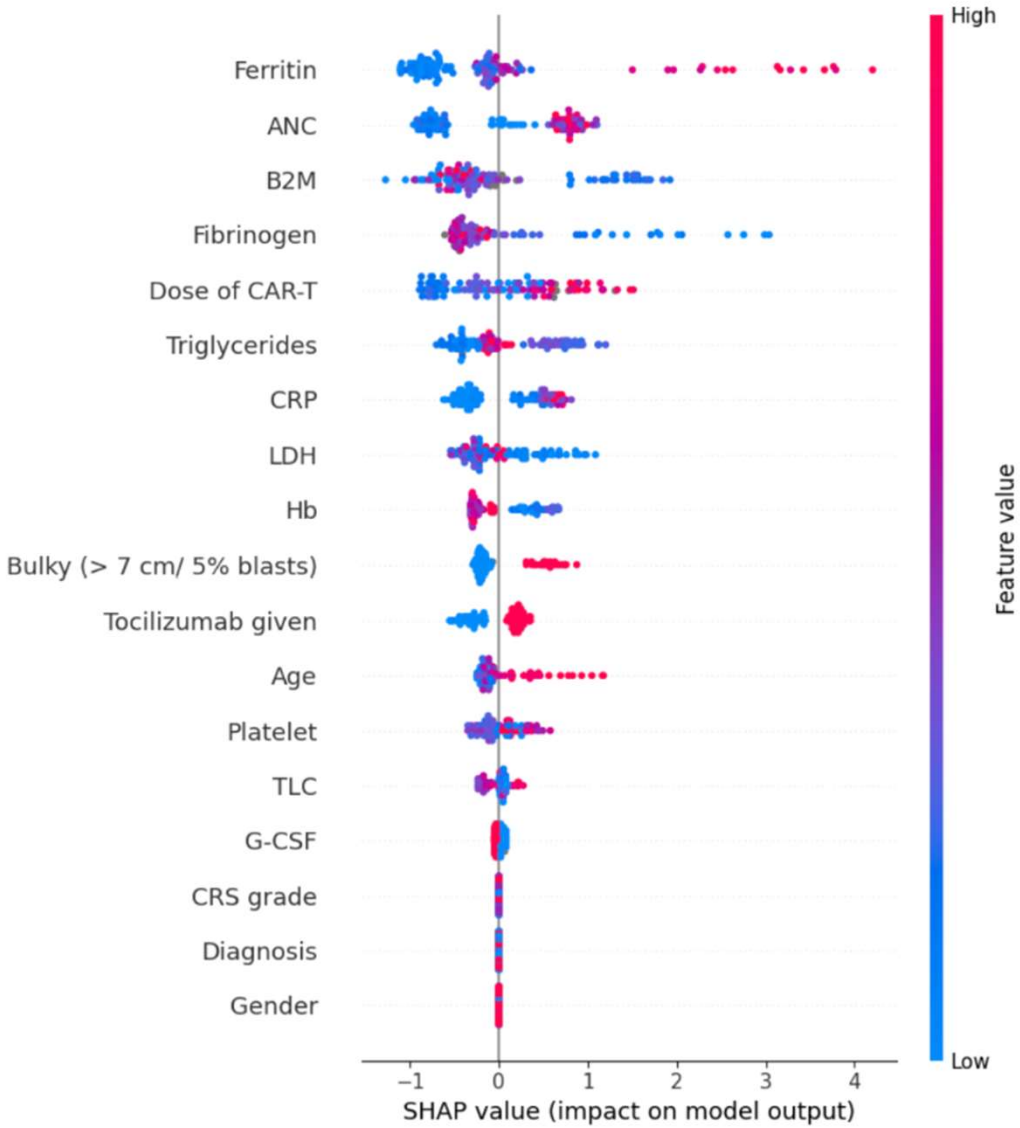
Code template

- Data preparation,
- Model (GPU-enabled) set to train and test.
- Confusion matrix, precision,recall, accuracies, F1 Score etc.
- XGBoost feature importance.
- Shapely analysis of individual decisions as well as summary view.
- Feature correlation heatmap.

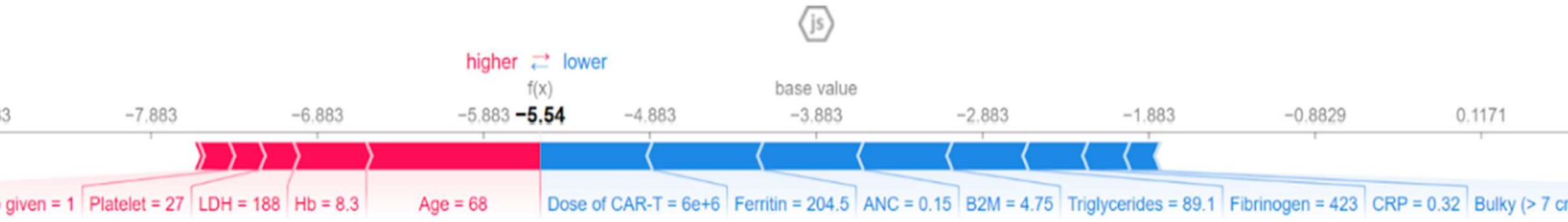
XGBoost feature importance summary

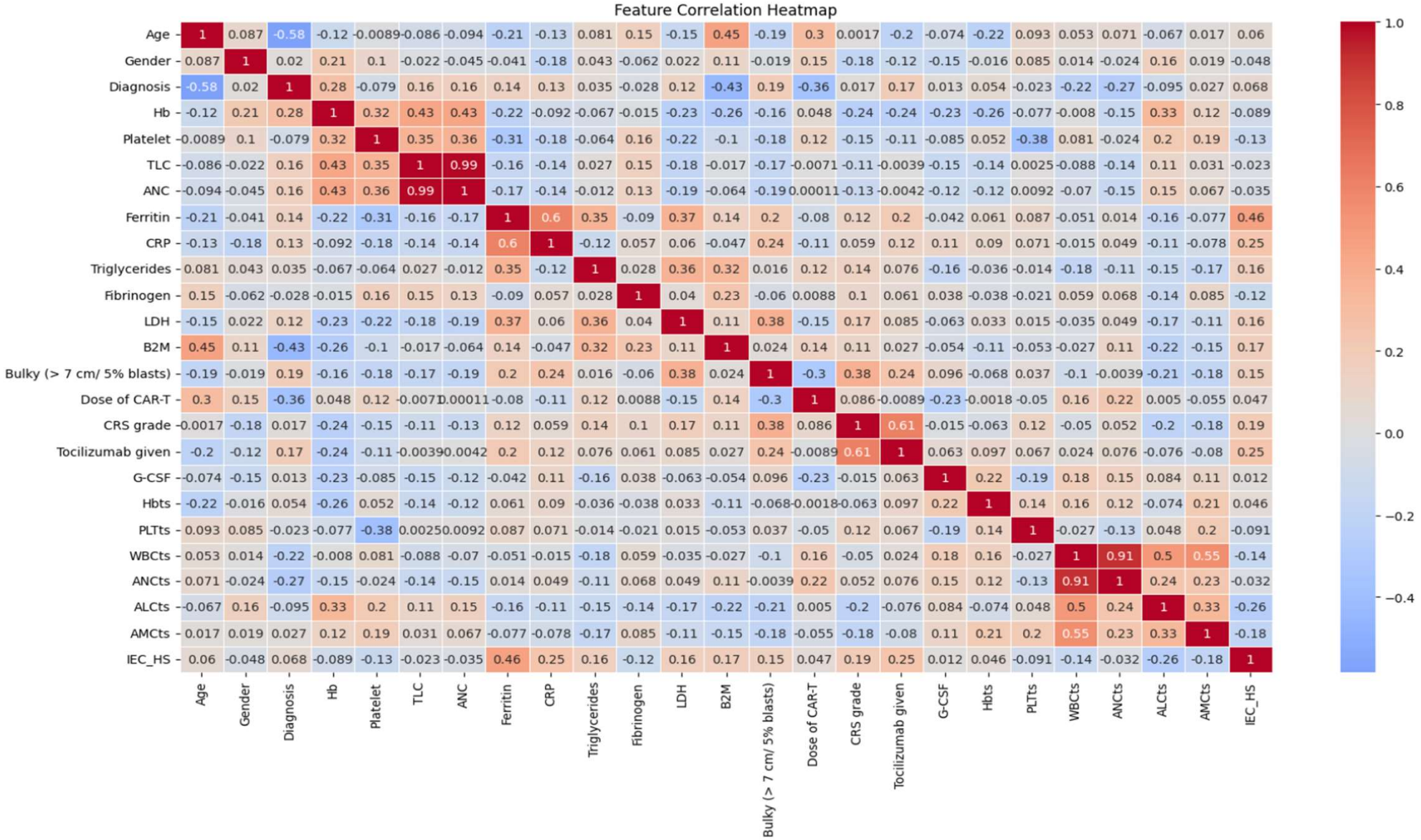


SHapely analysis summary



Influencing individual decisions




Feature Correlation Heatmap
Age 1 0.087 -0.58 -0.12 -0.0089 -0.086 -0.094 -0.21 -0.13 0.081 0.15 -0.15 0.45 -0.19 0.3 0.0017 -0.2 -0.074 -0.22 0.093 0.053 0.071 -0.067 0.017 0.06
Gender 0.087 1 0.02 0.21 0.1 -0.022 -0.045 -0.041 -0.18 0.043 -0.062 0.022 0.11 -0.019 0.15 -0.18 -0.12 -0.15 -0.016 0.085 0.014 -0.024 0.16 0.019 -0.048
Diagnosis -0.58 0.02 1 0.28 -0.079 0.16 0.16 0.14 0.13 0.035 -0.028 0.12 -0.43 0.19 -0.36 0.017 0.17 0.013 0.054 -0.023 -0.22 -0.27 -0.095 0.027 0.068
Hb -0.12 0.21 0.28 1 0.32 0.43 0.43 -0.22 -0.092 -0.067 -0.015 -0.23 -0.26 -0.16 0.048 -0.24 -0.24 -0.23 -0.26 -0.077 -0.008 -0.15 0.33 0.12 -0.089
Platelet -0.0089 0.1 -0.079 0.32 1 0.35 0.36 -0.31 -0.18 -0.064 0.16 -0.22 -0.1 -0.18 0.12 -0.15 -0.11 -0.085 0.052 -0.38 0.081 -0.024 0.2 0.19 -0.13
TLC -0.086 -0.022 0.16 0.43 0.35 1 0.99 -0.16 -0.14 0.027 0.15 -0.18 -0.017 -0.17 -0.0071 -0.11 -0.0039 -0.15 -0.14 0.0025 -0.088 -0.14 0.11 0.031 -0.023
ANC -0.094 -0.045 0.16 0.43 0.36 0.99 1 -0.17 -0.14 -0.012 0.13 -0.19 -0.064 -0.19 0.00011 -0.13 -0.0042 -0.12 -0.12 0.0092 -0.07 -0.15 0.15 0.067 -0.035
Ferritin -0.21 -0.041 0.14 -0.22 -0.31 -0.16 -0.17 1 0.6 0.35 -0.09 0.37 0.14 0.2 -0.08 0.12 0.2 -0.042 0.061 0.087 -0.051 0.014 -0.16 -0.077 0.46
CRP -0.13 -0.18 0.13 -0.092 -0.18 -0.14 -0.14 0.6 1 -0.12 0.057 0.06 -0.047 0.24 -0.11 0.059 0.12 0.11 0.09 0.071 -0.015 0.049 -0.11 -0.078 0.25
Triglycerides 0.081 0.043 0.035 -0.067 -0.064 0.027 -0.012 0.35 -0.12 1 0.028 0.36 0.32 0.016 0.12 0.14 0.076 -0.16 -0.036 -0.014 -0.18 -0.11 -0.15 -0.17 0.16
Fibrinogen 0.15 -0.062 -0.028 -0.015 0.16 0.15 0.13 -0.09 0.057 0.028 1 0.04 0.23 -0.06 0.0088 0.1 0.061 0.038 -0.038 -0.021 0.059 0.068 -0.14 0.085 -0.12
LDH -0.15 0.022 0.12 -0.23 -0.22 -0.18 -0.19 0.37 0.06 0.36 0.04 1 0.11 0.38 -0.15 0.17 0.085 -0.063 0.033 0.015 -0.035 0.049 -0.17 -0.11 0.16
B2M 0.45 0.11 -0.43 -0.26 -0.1 -0.017 -0.064 0.14 -0.047 0.32 0.23 0.11 1 0.024 0.14 0.11 0.027 -0.054 -0.11 -0.053 -0.027 0.11 -0.22 -0.15 0.17
Bulky (> 7 cm/ 5% blasts) -0.19 -0.019 0.19 -0.16 -0.18 -0.17 -0.19 0.2 0.24 0.016 -0.06 0.38 0.024 1 -0.3 0.38 0.24 0.096 -0.068 0.037 -0.1 -0.0039 -0.21 -0.18 0.15
Dose of CAR-T 0.3 0.15 -0.36 0.048 0.12 -0.0071 0.00011 -0.08 -0.11 0.12 0.0088 -0.15 0.14 -0.3 1 0.086 -0.0089 -0.23 -0.0018 -0.05 0.16 0.22 0.005 -0.055 0.047
CRS grade -0.0017 -0.18 0.017 -0.24 -0.15 -0.11 -0.13 0.12 0.059 0.14 0.1 0.17 0.11 0.38 0.086 1 0.61 -0.015 -0.063 0.12 -0.05 0.052 -0.2 -0.18 0.19
Tocilizumab given -0.2 -0.12 0.17 -0.24 -0.11 -0.0039 0.0042 0.2 0.12 0.076 0.061 0.085 0.027 0.24 -0.0089 0.61 1 0.063 0.097 0.067 0.024 0.076 -0.076 -0.08 0.25
G-CSF -0.074 -0.15 0.013 -0.23 -0.085 -0.15 -0.12 -0.042 0.11 -0.16 0.038 -0.063 -0.054 0.096 -0.23 -0.015 0.063 1 0.22 -0.19 0.18 0.15 0.084 0.11 0.012
Hbts -0.22 -0.016 0.054 -0.26 0.052 -0.14 -0.12 0.061 0.09 -0.036 -0.038 0.033 -0.11 -0.068 -0.0018 -0.063 0.097 0.22 1 0.14 0.16 0.12 -0.074 0.21 0.046
PLTts 0.093 0.085 -0.023 -0.077 -0.38 0.0025 0.0092 0.087 0.071 -0.014 -0.021 0.015 -0.053 0.037 -0.05 0.12 0.067 -0.19 0.14 1 -0.027 -0.13 0.048 0.2 -0.091
WBCts 0.053 0.014 -0.22 -0.008 0.081 -0.088 -0.07 -0.051 -0.015 -0.18 0.059 -0.035 -0.027 -0.1 0.16 -0.05 0.024 0.18 0.16 -0.027 1 0.91 0.5 0.55 -0.14
ANCts 0.071 -0.024 -0.27 -0.15 -0.024 -0.14 -0.15 0.014 0.049 -0.11 0.068 0.049 0.11 -0.0039 0.22 0.052 0.076 0.15 0.12 -0.13 0.91 1 0.24 0.23 -0.032
ALCts -0.067 0.16 -0.095 0.33 0.2 0.11 0.15 -0.16 -0.11 -0.15 -0.14 -0.17 -0.22 -0.21 0.005 -0.2 -0.076 0.084 -0.074 0.048 0.5 0.24 1 0.33 -0.26
AMCts 0.017 0.019 0.027 0.12 0.19 0.031 0.067 -0.077 -0.078 -0.17 0.085 -0.11 -0.15 -0.18 -0.055 -0.18 -0.08 0.11 0.21 0.2 0.55 0.23 0.33 1 -0.18
IEC_HS 0.06 -0.048 0.068 -0.089 -0.13 -0.023 -0.035 0.46 0.25 0.16 -0.12 0.16 0.17 0.15 0.047 0.19 0.25 0.012 0.046 -0.091 -0.14 -0.032 -0.26 -0.18 1
Age
Gender
Diagnosis
Hb
Platelet
TLC
ANC
Ferritin
CRP
Triglycerides
Fibrinogen
LDH
B2M
Bulky (> 7 cm/ 5% blasts)
Dose of CAR-T
CRS grade
Tocilizumab given
G-CSF
Hbts
PLTts
WBCts
ANCts
ALCts
AMCts
IEC_HS
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Further experiments (that didn’t work)

- We tried anomaly detection (XGBOD), synthetic sample generation (SMOTE) etc.
- We also played around with using SVM and CatBoost.
- Experimenting with different train-test splits and further model parameter tweaking- none it of it worked.

Why and how to rebalance ?

- This is due to less number of samples and class imbalance of 1:5.5.
- Thus we rebalance the data.
- Random selection – Biased/not conserving original data pattern.
- Submodlib- Nope. Still lacking strong reasoning. And bad accuracies.

Balanced dataset

- We select 18 IEC_HS=1 cases by matching Age interval-Gender-Diagnosis triplets with the 18 IEC_HS=0 cases.
- These 3 features are selected since they are fundamental to the patient and do not significantly drive the prediction.
- Statistically, these do not have class-wise significant differences.

Bell Distribution test (and CLT is applicable here)

Age	Group 0: Non-Normal (p=0.0001), Group 1: Non-Normal (p=0.0477)
Hb	Group 0: Normal (p=0.1110), Group 1: Normal (p=0.1122)
Platelet	Group 0: Non-Normal (p=0.0000), Group 1: Normal (p=0.0819)
TLC	Group 0: Non-Normal (p=0.0000), Group 1: Normal (p=0.0834)
ANC	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0464)
Ferritin	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0000)
CRP	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0000)
Triglycerides	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0004)
Fibrinogen	Group 0: Non-Normal (p=0.0079), Group 1: Normal (p=0.2735)
LDH	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0000)
B2M	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0002)
Dose of CAR-T	Group 0: Non-Normal (p=0.0000), Group 1: Normal (p=0.0532)
CRS grade	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0001)
Hbts	Group 0: Non-Normal (p=0.0000), Group 1: Normal (p=0.4177)
PLTts	Group 0: Normal (p=0.0800), Group 1: Normal (p=0.1977)
WBCts	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0297)
ANCts	Group 0: Non-Normal (p=0.0000), Group 1: Normal (p=0.3896)
ALCts	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0003)
AMCts	Group 0: Non-Normal (p=0.0000), Group 1: Non-Normal (p=0.0000)

Welch T-test

```
Ferritin                        | t-test | stat =  -2.2024, p-value = 0.0417
CRS grade                       | t-test | stat =  -2.1738, p-value = 0.0396
Bulky (> 7 cm/ 5% blasts) | chi2   | stat =   4.5675, p-value = 0.0326
WBCts                           | t-test | stat =   2.2931, p-value = 0.0270
Tocilizumab given               | chi2   | stat =   6.4747, p-value = 0.0109
ALCts                           | t-test | stat =   3.8678, p-value = 0.0005
```

```
Age                             | t-test | stat =   0.2677, p-value = 0.7915
```

Chi2 test and matrix for Gender and Diagnosis

```
Feature: Gender
Contingency Table (Observed counts):
IEC_HS   0   1
Gender
0       19   4
1       80  14

Chi2 Statistic: 0.0000
Degrees of Freedom: 1
P-value: 1.0000
Expected Frequencies:
IEC_HS          0          1
Gender
0       19.461538   3.538462
1       79.538462  14.461538
```

```
Feature: Diagnosis
Contingency Table (Observed counts):
IEC_HS      0   1
Diagnosis
0          53   7
1          46  11

Chi2 Statistic: 0.7872
Degrees of Freedom: 1
P-value: 0.3749
Expected Frequencies:
IEC_HS             0         1
Diagnosis
0          50.769231  9.230769
1          48.230769  8.769231
```

First pass in balanced dataset

- Confusion matrix is as: TP:11, TN:12, FP:6, FN:7
- Overall accuracy- 63.9%, Precision- 64.7%, Recall- 61.1%, F1 Score: 62.9%
- We need to narrow down our feature space !

Optimal feature set

- We find the optimal set of features for the balanced dataset by-
- Greedy Elimination:- Start with all 24 features and dynamically drop the least important feature in each iteration (a full LOOCV loop).
- Along with some manual tweaking, we always get 3,4 or 5 feature subsets with best performance.
- Thus we perform an Exhaustive search for the optimal 3/4/5 feature subset in the whole feature space.

Best feature subset

- Features: ('Hb', 'B2M', 'CRS grade', 'Tocilizumab given', 'G-CSF') → Accuracy: 77.78%
- So we train our model on the balanced 36 (18+18) and test on remaining 81 cases (All IEC_HS=0).
- We do this by: Selecting all features and selecting only optimal 5 feature set.
- Both give the same result: 55 TP, 26 FN i.e. Accuracy 67.8%.

(Rough work)

- 1. Features: ('Hb', 'B2M', 'CRS grade', 'Tocilizumab given', 'G-CSF') → Accuracy: 0.7778
- 2. Features: ('Hb', 'Platelet', 'B2M', 'Tocilizumab given', 'G-CSF') → Accuracy: 0.7500
- 3. Features: ('Hb', 'TLC', 'Fibrinogen', 'B2M', 'Tocilizumab given') → Accuracy: 0.7500
- 4. Features: ('Hb', 'TLC', 'B2M', 'CRS grade', 'Tocilizumab given') → Accuracy: 0.7500
- 5. Features: ('Hb', 'Fibrinogen', 'B2M', 'CRS grade', 'Tocilizumab given') → Accuracy: 0.7500

Best 4

- Features: ('Hb', 'B2M', 'CRS grade', 'Tocilizumab given') → Accuracy: 0.7778
- 2. Features: ('Hb', 'B2M', 'Tocilizumab given', 'G-CSF') → Accuracy: 0.7778
- 3. Features: ('Hb', 'TLC', 'B2M', 'Tocilizumab given') → Accuracy: 0.7500
- 4. Features: ('Hb', 'Fibrinogen', 'B2M', 'Tocilizumab given') → Accuracy: 0.7500
- 5. Features: ('Platelet', 'TLC', 'B2M', 'Tocilizumab given') → Accuracy: 0.7500

Optimal features